

A 3-connected graph in which no positive edge lengths make every geodesic cycle peripheral: the Kleetope of K_4 and Problem 3 of Georgakopoulos and Sprüssel

github.com/graph-theory-AI

Machine-generated note

17 September 2026

Abstract

Georgakopoulos and Sprüssel proved that for every finite graph G and every assignment ℓ of positive lengths to its edges the ℓ -geodesic cycles of G generate its cycle space, and asked (Problem 3 of [1], also listed in the Open Problem Garden) whether every 3-connected finite graph admits an ℓ for which every ℓ -geodesic cycle is *peripheral*, i.e. induced and non-separating; a positive answer would give a new proof of Tutte’s theorem that the peripheral cycles of a 3-connected graph generate its cycle space. We show that the answer is negative. Let G be the planar 3-connected graph on eight vertices obtained from K_4 by adding a new vertex inside each of its four triangles and joining it to the three vertices of that triangle (the Kleetope of the tetrahedron). For every $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ some ℓ -geodesic cycle of G is induced but separating. The proof is a counting argument: G has exactly twelve peripheral cycles, all triangles, while its cycle space has dimension eleven; once shortest paths are made unique by a perturbation, each edge of the K_4 that does not realise the distance between its ends destroys the geodesicity of one of the two peripheral triangles through it, so at most one such edge exists, and a triangle of the K_4 avoiding it is geodesic yet separating.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

All graphs are finite and simple. For a graph H and an assignment $\ell: E(H) \rightarrow \mathbb{R}_{>0}$ of positive *lengths* to its edges, the length of a subgraph F (in particular of a path or a cycle) is $\ell(F) = \sum_{e \in E(F)} \ell(e)$, and $d_\ell(u, v)$ denotes the least length of a u - v path; on a connected graph d_ℓ is a metric on $V(H)$. Two distinct vertices x, y of a cycle C split C into two x - y paths, the *arcs* of C between x and y . Georgakopoulos and Sprüssel [1, §3.1] define, for finite graphs:

“A cycle C in G is ℓ -geodetic, if for any $x, y \in V(C)$ there is no x - y path in G of length strictly less than that of each of the two x - y paths on C .”

Following the Open Problem Garden entry [2] we say ℓ -geodesic for ℓ -geodetic. Two points of this definition matter below: it quantifies over pairs of *vertices* of C (not over all points of C regarded as a topological circle, which is the notion [1, §2] uses for infinite graphs), and the inequality is *strict*, so a path of length exactly equal to the shorter arc does not destroy geodesicity. Equivalently, C is ℓ -geodesic if and only if $d_\ell(x, y) = d_C(x, y)$ for all $x, y \in V(C)$, where $d_C(x, y)$ is the length of the shorter arc.

The cycle space $\mathcal{C}(H)$ is the $\mathbb{F}_2$ -vector space spanned by the edge sets of the cycles of H inside $\mathbb{F}_2^{E(H)}$; for connected H it has dimension $|E(H)| - |V(H)| + 1$ [4, §1.9]. A set of cycles *generates* $\mathcal{C}(H)$ if their edge sets span it. The second definition, from [1, §5], is:

“Call a cycle in a finite graph peripheral, if it is induced and non-separating.”

Here *induced* means that C has no chord (every edge of H between two vertices of C is an edge of C), and *non-separating* means that $H - V(C)$ is connected. This is the notion of Tutte [3], whose classical theorem states that the peripheral cycles of a 3-connected graph generate its cycle space.

Georgakopoulos and Sprüssel [1, Theorem 3.1] proved that for every finite graph H and every ℓ the ℓ -geodesic cycles generate $\mathcal{C}(H)$ (Theorem 3.1 below), and in the concluding section of [1] they posed the following problem, whose positive answer would combine with that theorem into a new proof of Tutte’s theorem. In the wording of the Open Problem Garden [2], which attributes it to them:

Problem 1.1 ([1, Problem 3]; [2]). If G is a 3-connected finite graph, is there an assignment of lengths $\ell: E(G) \rightarrow \mathbb{R}^+$ to the edges of G , such that every ℓ -geodesic cycle is peripheral?

The source’s own wording is “If G is a 3-connected finite graph, is there an assignment of lengths ℓ to the edges of G , such that every ℓ -geodesic cycle is peripheral? We were not able to give an answer to this problem” [1, §5]; in [1] every assignment of positive edge lengths to a finite graph is admissible, so the two formulations agree. Neither the referee report reproduced in Appendix B nor the literature review on the catalogue page of the problem found a published answer in either direction; the closest work the latter located, Hamann’s study of generating sets of the cycle space of 3-connected planar graphs [5], does not concern edge lengths or geodesic cycles.

We answer Problem 1.1 negatively with an explicit graph on eight vertices. Only the two definitions above enter; in particular *both* halves of “peripheral” are needed (Remark 4.1), and the argument says nothing against Tutte’s theorem itself, only against this route to it.

2 The graph and the main theorem

Definition 2.1 (The Kleetope of K_4). Let $B = \{b_1, b_2, b_3, b_4\}$ and $S = \{s_1, s_2, s_3, s_4\}$ be disjoint four-element sets and let G be the graph with

$$V(G) = B \cup S, \quad E(G) = \{b_i b_j : 1 \leq i < j \leq 4\} \cup \{s_i b_j : i \neq j\}.$$

Explicitly, the 18 edges are the six *core* edges $b_1 b_2, b_1 b_3, b_1 b_4, b_2 b_3, b_2 b_4, b_3 b_4$ and the twelve *spoke* edges

$$s_1 b_2, s_1 b_3, s_1 b_4, s_2 b_1, s_2 b_3, s_2 b_4, s_3 b_1, s_3 b_2, s_3 b_4, s_4 b_1, s_4 b_2, s_4 b_3.$$

Thus B induces a K_4 , S is independent, and s_i is adjacent to the three vertices of $B \setminus \{b_i\}$. Geometrically, G is a tetrahedron with a new vertex stacked in each face and joined to that face’s three corners; this is the Kleetope of K_4 , the graph of the triakis tetrahedron. For $i \in \{1, 2, 3, 4\}$ write

$$\begin{aligned} T_i &= G[B \setminus \{b_i\}] && \text{(the four core triangles),} \\ P_{i,jk} &= G[\{s_i, b_j, b_k\}] \quad (j < k, i \notin \{j, k\}) && \text{(the twelve spoke triangles).} \end{aligned}$$

Theorem 2.2 (Main theorem). Let G be the graph of Definition 2.1. Then G is a planar 3-connected graph on 8 vertices with 18 edges, and for every assignment $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ some ℓ -geodesic cycle of G is not peripheral.

Corollary 2.3. The answer to Problem 1.1 is negative, already for planar 3-connected graphs on eight vertices.

Proof. Immediate from [Theorem 2.2](#): G is 3-connected and finite, and no ℓ makes every ℓ -geodesic cycle peripheral. $\square$

The structure of G is recorded in the following lemma.

Lemma 2.4 (Structure of G). *Let G be the graph of [Definition 2.1](#).*

- (a) G is planar, 3-connected, and $\dim_{\mathbb{F}_2} \mathcal{C}(G) = 18 - 8 + 1 = 11$.
- (b) Every induced cycle of G is a triangle. The triangles of G are exactly the four core triangles T_i and the twelve spoke triangles $P_{i;jk}$, sixteen in all.
- (c) No core triangle is peripheral: s_i is an isolated vertex of $G - V(T_i)$, which has a second, nonempty component. Every spoke triangle is peripheral. Hence G has exactly twelve peripheral cycles, the $P_{i;jk}$.
- (d) Each spoke triangle $P_{i;jk}$ contains exactly one core edge, namely $b_j b_k$. Each core edge $b_j b_k$ lies in exactly two peripheral cycles, $P_{p;jk}$ and $P_{q;jk}$ where $\{p, q\} = \{1, 2, 3, 4\} \setminus \{j, k\}$.
- (e) Each core edge lies in exactly two of the four core triangles; so for every core edge e there is a core triangle T_i with $e \notin E(T_i)$.

Idea of proof. Deleting at most two vertices leaves at least two (adjacent) vertices of the clique B , and every surviving s_i keeps one of its three neighbours in B . An induced cycle of length at least four cannot contain an s_i , since the two cycle-neighbours of s_i lie in the clique B and would be joined by a chord, and cannot lie inside B either. The rest is inspection of the sixteen triangles. [\[full proof\]](#)

3 Three lemmas on edge lengths

The first statement is a theorem of Georgakopoulos and Sprüssel; the original writeup ([Section 5](#)) rederived it as its “Fact 1”. We include a proof in [Appendix A](#) only to keep the note self-contained and because the referee asked for one step of the standard argument to be spelled out.

Theorem 3.1 (Georgakopoulos–Sprüssel [[1](#), Theorem 3.1]). *Let H be a finite graph and $\ell: E(H) \rightarrow \mathbb{R}_{>0}$. Every cycle C of H is a sum, in $\mathcal{C}(H)$, of ℓ -geodesic cycles each of length at most $\ell(C)$. In particular the ℓ -geodesic cycles generate $\mathcal{C}(H)$.*

Idea of proof. If C is not geodesic, some path P between two of its vertices is shorter than both arcs. Cutting P at its visits to C and using the triangle inequality for the shorter-arc metric on $V(C)$, one of the resulting segments Q is a path with distinct ends u, v on C , all interior vertices off C , and $\ell(Q)$ smaller than both u - v arcs A, B . Then $C_1 = A \cup Q$ and $C_2 = B \cup Q$ are cycles, $E(C) = E(C_1) \triangle E(C_2)$ and both are strictly shorter than C ; induct on the finitely many cycle lengths. [\[full proof\]](#)

Lemma 3.2 (Perturbation to unique shortest paths). *Let H be a finite graph and suppose $\ell_0: E(H) \rightarrow \mathbb{R}_{>0}$ is such that every ℓ_0 -geodesic cycle of H is peripheral. Then there is $\ell: E(H) \rightarrow \mathbb{R}_{>0}$ such that every ℓ -geodesic cycle of H is peripheral and, for every two vertices u, v of H , any two u - v paths of minimum ℓ -length coincide.*

Idea of proof. Each of the finitely many non-peripheral cycles is not ℓ_0 -geodesic, which is witnessed by two strict linear inequalities in ℓ_0 ; together with positivity they hold on an open neighbourhood U of ℓ_0 in $\mathbb{R}^{E(H)}$. Ties $\ell(P) = \ell(Q)$ between distinct paths with the same ends are finitely many proper hyperplanes, whose union misses some point of U . The lemma needs only that non-peripheral cycles stay non-geodesic; it does not claim that peripheral cycles stay geodesic. [\[full proof\]](#)

Call an edge uv of H *tight* for ℓ if $\ell(uv) = d_\ell(u, v)$. Since the edge itself is a u - v path, always $d_\ell(u, v) \leq \ell(uv)$, so a non-tight edge is strictly longer than the distance between its ends.

Lemma 3.3 (Tight edges). (i) *In any finite graph H with any $\ell: E(H) \rightarrow \mathbb{R}_{>0}$, a triangle all of whose three edges are tight is ℓ -geodesic.*

- (ii) Let G be the graph of [Definition 2.1](#) and let $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ be such that any two shortest paths with the same ends coincide. If a core edge $b_j b_k$ is not tight, then at most one of the two spoke triangles $P_{p;jk}, P_{q;jk}$ containing it is ℓ -geodesic.

Idea of proof. (i): for two vertices of the triangle, the edge joining them is one of the two arcs and has length $d_\ell(x, y)$, so no path is strictly shorter than both arcs. (ii): if $P_{p;jk}$ is geodesic and $d_\ell(b_j, b_k) < \ell(b_j b_k)$, then the arc $b_j s_p b_k$ must be a shortest b_j - b_k path; the same for $P_{q;jk}$ would give two distinct shortest paths. [\[full proof\]](#)

Idea of proof (of [Theorem 2.2](#)). Suppose some ℓ_0 makes every geodesic cycle peripheral; by [Lemma 3.2](#) we may assume shortest paths are unique. By [Lemma 2.4\(c\)](#) the geodesic cycles are then among the twelve spoke triangles, and by [Theorem 3.1](#) they span the 11-dimensional cycle space, so at most one spoke triangle is non-geodesic. By [Lemma 3.3\(ii\)](#) and [Lemma 2.4\(d\)](#), every non-tight core edge lies in a non-geodesic spoke triangle, and distinct core edges give distinct triangles; so at most one core edge is non-tight. A core triangle avoiding it ([Lemma 2.4\(e\)](#)) has all edges tight, hence is geodesic by [Lemma 3.3\(i\)](#), but it is separating by [Lemma 2.4\(c\)](#): a geodesic non-peripheral cycle, contradiction. [\[full proof\]](#)

4 Remarks

Remark 4.1 (The definitions are load-bearing). The refutation uses exactly the definitions of [1] quoted in the introduction, and it depends on all of their parts. (a) *Both halves of “peripheral”*. The core triangles T_i are induced but separating; the argument fails if either half is dropped, and this is not an artefact of the proof: the referee’s exact decision procedure (below) finds that the same graph G *does* admit positive lengths making every geodesic cycle non-separating (there are 120 non-separating cycles), and also positive lengths making every geodesic cycle induced (the 16 triangles). Any restatement of [Theorem 2.2](#) must therefore keep “induced and non-separating”. (b) *Vertices, strict inequality*. Geodesicity is tested on pairs of vertices with a strict inequality, which is the finite-graph definition of [1, §3.1] and the one quoted by [2]. [Lemma 3.3\(i\)](#) is a vertex-level statement; under the point-level definition that [1, §2] uses for topological circles in infinite graphs it would need re-examination. The referee judged the vertex reading to be the intended one for [Problem 1.1](#), so this is a scope note rather than a gap.

Remark 4.2 (The proof does not locate the geodesic non-peripheral cycle). [Added in the rewriting stage, [Section 5](#).] The contradiction in the proof of [Theorem 2.2](#) is reached for a *perturbed* assignment ℓ and exhibits a geodesic core triangle for that ℓ ; since [Lemma 3.2](#) may change which cycles are geodesic, this does not show that a core triangle is geodesic for the original assignment, and indeed it need not be. For instance, give every core edge length 10 and every spoke edge length 1. Then $d_\ell(b_j, b_k) = 2 < 10$, so no core triangle is ℓ -geodesic (and every spoke triangle is), while the 4-cycle $b_1 s_3 b_2 s_4$ is ℓ -geodesic (each of its vertex pairs is at distance 1 or 2, equal to the shorter arc) and has the chord $b_1 b_2$, so it is not peripheral. A direct check over all 239 cycles of G finds ten ℓ -geodesic non-peripheral cycles for this ℓ , all of length 4 or 6, and no geodesic core triangle. [Theorem 2.2](#) is therefore stated, as in the original writeup, purely existentially.

Remark 4.3 (Referee’s computational checks). The adversarial referee ([Appendix B](#); scripts in `verification_astra/scripts/geodesic_cycles_and_tuttes_theorem/`) verified [Lemma 2.4](#) by brute force with `networkx`: 8 vertices, 18 edges, degrees 6 on B and 3 on S , vertex connectivity exactly 3, planar, $\dim \mathcal{C}(G) = 11$; 239 simple cycles (16 triangles, 33 of length 4, 60 of length 5, 76 of length 6, 48 of length 7, 6 of length 8), of which exactly 16 are induced, all triangles, and exactly 12 are peripheral, namely the $P_{i;jk}$; each spoke triangle contains one core edge, each core edge and each spoke edge lies in exactly two peripheral triangles, and the fewest edges of K_4 meeting all four of its triangles is 2. Over $\mathbb{F}_2$ the twelve peripheral cycles have rank 11, and every choice of ten of them has rank exactly 10, so two non-geodesic spoke triangles already break generation (the proof needs only the weaker fact that ten vectors span at most ten dimensions).

Independently of the argument, the referee encoded the statement “there exists $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ such that every ℓ -geodesic cycle is peripheral” exactly in linear real arithmetic (QF_LRA) for the SMT solver z3: length variables $\ell_e > 0$, distance variables constrained by the Bellman characterisation of shortest-path distances, and for each of the 227 non-peripheral cycles a disjunction over vertex pairs x, y of the cycle of “ $d(x, y)$ is strictly less than both arcs”. The solver returned unsat, which, since linear real arithmetic is decided exactly, is a machine proof of [Theorem 2.2](#) not relying on the human-readable argument. To rule out a vacuous encoding, the same code returned sat on K_4 , K_5 , the cube Q_3 , the triangular prism, the octahedron and the wheel W_4 , and in each case the rational model was re-checked with Dijkstra’s algorithm and found to have no non-peripheral geodesic cycle (for K_4 on vertices 0, 1, 2, 3: $\ell(01) = \ell(12) = \ell(13) = 1/8$, $\ell(02) = \ell(03) = \ell(23) = 3/8$). On G itself the encoding returned sat under each of the two weakened readings of “peripheral” mentioned in [Remark 4.1](#). Finally, 40 000 random assignments from four distributions never produced fewer than 2 non-peripheral geodesic cycles, and simulated annealing on log-lengths got stuck at 4.

Remark 4.4 (Minimality, novelty, scope). Minimality of the example is not claimed: eight vertices is small, but no smaller counterexample has been excluded. The referee’s controls show that K_4 , K_5 , Q_3 , the prism, the octahedron and W_4 are not counterexamples, which is consistent with, but far from a proof of, minimality. Neither the model nor the referee found this example or any answer to [Problem 1.1](#) in the literature, and the catalogue page’s review found none either; but the topic is niche and “not found” is weaker than “not there”. The authors of [1] should be asked whether the example was known to them; if it was folklore, the contribution of this note reduces to a written proof. [Theorem 2.2](#) blocks the proposed route to a new proof of Tutte’s theorem via geodesic cycles; it says nothing against Tutte’s theorem. It leaves open the natural weakening (which 3-connected graphs *do* admit an ℓ making every ℓ -geodesic cycle peripheral? the six control graphs above do) and the infinite version of the problem discussed in [1].

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-6-astra (reasoning effort *max*) in a single autonomous pass on 2026-09-16, as part of a sweep over the problems of the Open Problem Garden (catalog id *geodesic_cycles_and_tuttes_theorem*, leg *attacks_opg*; original writeup in *attacks_opg/geodesic_cycles_and_tuttes_theorem/output.md*). The model rated its claim “disproved” with high confidence and flagged that it had not checked novelty. (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17: it retrieved the published source [1] and checked the definitions of ℓ -geodetic and peripheral cycles, [Theorem 3.1](#) and [Problem 3](#) verbatim; re-derived all seventeen steps of the writeup and labelled each VALID; verified [Lemma 2.4](#) by exhaustive computation; and proved [Theorem 2.2](#) independently with an exact z3 decision procedure, with positive controls ([Remark 4.3](#)). Its verdict was CONFIRMED, with the reservations recorded in [Remarks 4.1](#) and [4.4](#). Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The text was reorganised with full proofs in [Appendix A](#). Deviations from the writeup, all declared here: the writeup’s “Fact 1” is attributed to Georgakopoulos and Sprüssel and stated as their [Theorem 3.1](#) (the writeup proved it without citation); its “Fact 2” is [Lemma 3.2](#); the two cosmetic gaps named by the referee were filled in [Appendix A](#), namely the existence of a shortcut path internally disjoint from the cycle in the proof of [Theorem 3.1](#) (the writeup’s “otherwise every excursion of P outside C could be replaced by a no-longer arc of C ”, made precise via the triangle inequality for the shorter-arc metric on $V(C)$), and the properness of the hyperplanes $\ell(P) = \ell(Q)$ in the proof of [Lemma 3.2](#) (distinct paths have distinct edge sets, and a finite union of hyperplanes has empty interior); the writeup’s “choose a core triangle avoiding [the exceptional edge]” is justified by the explicit count in [Lemma 2.4\(e\)](#); the verbatim definitions and the wording of [Problem 3](#) from [1], the name “Kleetope of K_4 ”, the explicit edge list, the reference [4] for the dimension of the cycle space, and the content of [Remarks 4.1](#), [4.3](#) and [4.4](#) were taken from

the referee report; the reference [5] was taken from the literature review on the catalogue page of the problem. One item is the rewriter's own: [Remark 4.2](#), which records that the argument does not identify the geodesic non-peripheral cycle for a given ℓ and gives an explicit assignment (core edges 10, spoke edges 1) for which no core triangle is geodesic, checked by enumerating all 239 cycles; it is a scope note and is not used in any proof. An earlier draft of this note had strengthened [Theorem 2.2](#) to assert that a core triangle is always geodesic; that strengthening is not in the writeup and is false by the same example, and it was removed. No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it for human review, not as a claim to a new resolution.

A Full proofs

Lemma 2.4 (Structure of G). *Let G be the graph of [Definition 2.1](#).*

- (a) G is planar, 3-connected, and $\dim_{\mathbb{F}_2} \mathcal{C}(G) = 18 - 8 + 1 = 11$.
- (b) Every induced cycle of G is a triangle. The triangles of G are exactly the four core triangles T_i and the twelve spoke triangles $P_{i;jk}$, sixteen in all.
- (c) No core triangle is peripheral: s_i is an isolated vertex of $G - V(T_i)$, which has a second, nonempty component. Every spoke triangle is peripheral. Hence G has exactly twelve peripheral cycles, the $P_{i;jk}$.
- (d) Each spoke triangle $P_{i;jk}$ contains exactly one core edge, namely $b_j b_k$. Each core edge $b_j b_k$ lies in exactly two peripheral cycles, $P_{p;jk}$ and $P_{q;jk}$ where $\{p, q\} = \{1, 2, 3, 4\} \setminus \{j, k\}$.
- (e) Each core edge lies in exactly two of the four core triangles; so for every core edge e there is a core triangle T_i with $e \notin E(T_i)$.

Proof. (a) G has 6 core and $4 \cdot 3 = 12$ spoke edges, 18 in all. For planarity, draw $K_4 = G[B]$ on the sphere as a tetrahedron; its four faces are bounded by the triangles $T_1, \dots, T_4$, and $N(s_i) = B \setminus \{b_i\} = V(T_i)$, so placing s_i inside the face bounded by T_i and joining it to the three corners gives a plane drawing (a check: $8 - 18 + 12 = 2$). For 3-connectivity let $X \subseteq V(G)$ with $|X| \leq 2$. Then $B \setminus X$ has at least two vertices and is a clique, hence connected in $G - X$. Every $s_i \notin X$ has three neighbours in B , of which at most two lie in X , so s_i has a neighbour in $B \setminus X$. Thus $G - X$ is connected, and since $|V(G)| = 8 > 3$, G is 3-connected (and not 4-connected, as $\deg s_i = 3$). Since G is connected, $\dim_{\mathbb{F}_2} \mathcal{C}(G) = |E(G)| - |V(G)| + 1 = 11$ [4, §1.9].

(b) Let C be an induced cycle of length at least 4. If some $s_i \in V(C)$, its two neighbours on C lie in $N(s_i) \subseteq B$, are distinct, and are therefore adjacent in G because B is a clique; as C has length at least 4 they are not consecutive on C , so the edge between them is a chord, contradicting that C is induced. Hence $V(C) \subseteq B$, so C is a 4-cycle through all of B , and its two diagonals are chords, again a contradiction. So every induced cycle is a triangle. Conversely every triangle is induced. Since S is independent, a triangle contains at most one vertex of S : the triangles inside B are the four T_i , and a triangle through s_i consists of s_i and two of its three neighbours $b_j, b_k \in B \setminus \{b_i\}$, giving $\binom{3}{2} = 3$ triangles $P_{i;jk}$ for each i , twelve in all.

(c) $G - V(T_i)$ has vertex set $\{b_i\} \cup S$. All three neighbours of s_i lie in $V(T_i)$, so s_i is isolated there, while b_i is adjacent to s_j for each $j \neq i$, so $\{b_i\} \cup (S \setminus \{s_i\})$ is a second component. Thus T_i is separating and not peripheral (it is induced, being a triangle). Now let $P = P_{i;jk}$ and $\{p, q\} = \{1, 2, 3, 4\} \setminus \{j, k\}$ (so $i \in \{p, q\}$). Then $G - V(P)$ has vertex set $\{b_p, b_q\} \cup (S \setminus \{s_i\})$; $b_p b_q$ is an edge, and each s_m with $m \neq i$ is non-adjacent to exactly one vertex of B , namely b_m , hence adjacent to at least one of b_p, b_q . So $G - V(P)$ is connected, and P is induced and non-separating, i.e. peripheral. By (b) the peripheral cycles, being induced, are triangles, so they are exactly the twelve $P_{i;jk}$.

(d) $E(P_{i;jk}) = \{s_i b_j, s_i b_k, b_j b_k\}$ contains exactly one core edge. The spoke triangles containing the core edge $b_j b_k$ are the $P_{i;jk}$ with $i \notin \{j, k\}$, i.e. $i \in \{p, q\}$: exactly two.

(e) The core triangles containing $b_j b_k$ are $G[B \setminus \{b_i\}]$ for $i \notin \{j, k\}$, exactly two of the four; the other two avoid $b_j b_k$. $\square$

Theorem 3.1 (Georgakopoulos–Sprüssel [1, Theorem 3.1]). *Let H be a finite graph and $\ell: E(H) \rightarrow \mathbb{R}_{>0}$. Every cycle C of H is a sum, in $\mathcal{C}(H)$, of ℓ -geodesic cycles each of length at most $\ell(C)$. In particular the ℓ -geodesic cycles generate $\mathcal{C}(H)$.*

Proof. Fix ℓ . For a cycle C and $x, y \in V(C)$ let $d_C(x, y)$ be 0 if $x = y$ and otherwise the length of the shorter of the two x - y arcs of C ; d_C is the shortest-path metric of the weighted graph $(C, \ell|_{E(C)})$, so it satisfies the triangle inequality on $V(C)$. In these terms C is ℓ -geodesic if and only if $\ell(P) \geq d_C(x, y)$ for all $x, y \in V(C)$ and all x - y paths P of H .

Claim. If C is not ℓ -geodesic, there are distinct $u, v \in V(C)$ and a u - v path Q of H all of whose interior vertices lie outside $V(C)$, with $\ell(Q) < d_C(u, v)$. [This is the step the referee asked to have spelled out; it is the “every excursion of P outside C could be replaced by a no-longer arc of C ” of the original writeup and of [1].] Indeed, let $x, y \in V(C)$ and an x - y path P satisfy $\ell(P) < d_C(x, y)$; necessarily $x \neq y$. Let $x = w_0, w_1, \dots, w_m = y$ be the vertices of P that lie on C , in their order along P ($m \geq 1$, all distinct because P is a path). The segments $P_t = P[w_t, w_{t+1}]$, $0 \leq t < m$, are paths with both ends on C and all interior vertices off C , and $\ell(P) = \sum_t \ell(P_t)$. If $\ell(P_t) \geq d_C(w_t, w_{t+1})$ for every t , then by the triangle inequality

$$\ell(P) = \sum_{t < m} \ell(P_t) \geq \sum_{t < m} d_C(w_t, w_{t+1}) \geq d_C(x, y),$$

contradicting the choice of P . So some segment $Q = P_t$ has $\ell(Q) < d_C(w_t, w_{t+1})$; put $u = w_t$, $v = w_{t+1}$. This proves the claim. Note that Q is not an edge of C , since for an edge $uv \in E(C)$ one has $d_C(u, v) \leq \ell(uv)$.

Now let A and B be the two u - v arcs of C , so $\ell(A), \ell(B) \geq d_C(u, v) > \ell(Q)$, and set $C_1 = A \cup Q$, $C_2 = B \cup Q$. Because the interior vertices of Q avoid $V(C)$ and Q is not an edge of C , Q is internally disjoint from A and from B and shares no edge with C (if Q has one edge it is a chord; if more, each of its edges has an end off C); hence C_1 and C_2 are cycles, and

$$E(C_1) \triangle E(C_2) = (E(A) \cup E(Q)) \triangle (E(B) \cup E(Q)) = E(A) \cup E(B) = E(C),$$

so $C = C_1 + C_2$ in $\mathcal{C}(H)$, while $\ell(C_1) = \ell(A) + \ell(Q) < \ell(A) + \ell(B) = \ell(C)$ and likewise $\ell(C_2) < \ell(C)$.

Finally, let $\lambda_1 < \lambda_2 < \dots < \lambda_N$ be the distinct values of ℓ on the finitely many cycles of H . We show by induction on k that every cycle C with $\ell(C) = \lambda_k$ is a sum of ℓ -geodesic cycles of length at most λ_k . If C is ℓ -geodesic there is nothing to prove; otherwise $C = C_1 + C_2$ with $\ell(C_1), \ell(C_2) < \lambda_k$, so $\ell(C_i) \in \{\lambda_1, \dots, \lambda_{k-1}\}$ (in particular $k \geq 2$, so a cycle of length λ_1 is always geodesic) and the induction hypothesis applies to C_1 and C_2 . Since $\mathcal{C}(H)$ is spanned by the edge sets of cycles, the ℓ -geodesic cycles generate it. $\square$

Lemma 3.2 (Perturbation to unique shortest paths). *Let H be a finite graph and suppose $\ell_0: E(H) \rightarrow \mathbb{R}_{>0}$ is such that every ℓ_0 -geodesic cycle of H is peripheral. Then there is $\ell: E(H) \rightarrow \mathbb{R}_{>0}$ such that every ℓ -geodesic cycle of H is peripheral and, for every two vertices u, v of H , any two u - v paths of minimum ℓ -length coincide.*

Proof. Identify assignments $\ell: E(H) \rightarrow \mathbb{R}$ with points of $\mathbb{R}^{E(H)}$; for a subgraph F , $\ell \mapsto \ell(F)$ is a linear form. Let $\mathcal{N}$ be the (finite) set of non-peripheral cycles of H . By hypothesis each $C \in \mathcal{N}$ is not ℓ_0 -geodesic, so we may fix $x_C, y_C \in V(C)$, an x_C - y_C path P_C , and the two arcs A_C, B_C of C between x_C and y_C , with $\ell_0(P_C) < \ell_0(A_C)$ and $\ell_0(P_C) < \ell_0(B_C)$. Let

$$U = \{\ell \in \mathbb{R}^{E(H)} : \ell(e) > 0 \ \forall e \in E(H); \ \ell(P_C) < \ell(A_C), \ell(P_C) < \ell(B_C) \ \forall C \in \mathcal{N}\}.$$

U is defined by finitely many strict linear inequalities, so it is open, and $\ell_0 \in U$. For every $\ell \in U$, every $C \in \mathcal{N}$ is not ℓ -geodesic (witnessed by P_C), i.e. every ℓ -geodesic cycle is peripheral.

For each pair $\{P, Q\}$ of distinct paths of H with the same two ends let $M_{P,Q} = \{\ell \in \mathbb{R}^{E(H)} : \ell(P) = \ell(Q)\}$. The defining equation $\sum_{e \in E(P)} \ell(e) - \sum_{e \in E(Q)} \ell(e) = 0$ has coefficient vector $\mathbf{1}_{E(P)} - \mathbf{1}_{E(Q)}$, which is nonzero because a path is determined by its edge set, so distinct paths have distinct edge sets. [Properness of these hyperplanes is the second point the referee asked to have made explicit.] Hence each $M_{P,Q}$ is a hyperplane, a closed set with empty interior, and there are finitely many of them. A finite union of closed sets with empty interior has empty interior (if U' is a nonempty open set and $M_1, \dots, M_r$ are closed with empty interior, then $U' \setminus M_1$ is nonempty and open, and inductively so is $U' \setminus (M_1 \cup \dots \cup M_r)$). Therefore $U \setminus \bigcup_{P,Q} M_{P,Q}$ is nonempty; pick ℓ in it.

Then ℓ is positive and every ℓ -geodesic cycle is peripheral. If R, R' are two u - v paths of minimum ℓ -length, then $\ell(R) = \ell(R')$, so $\ell \in M_{R,R'}$ unless $R = R'$; hence $R = R'$. $\square$

Lemma 3.3 (Tight edges). (i) *In any finite graph H with any $\ell: E(H) \rightarrow \mathbb{R}_{>0}$, a triangle all of whose three edges are tight is ℓ -geodesic.*

(ii) *Let G be the graph of Definition 2.1 and let $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ be such that any two shortest paths with the same ends coincide. If a core edge $b_j b_k$ is not tight, then at most one of the two spoke triangles $P_{p;jk}, P_{q;jk}$ containing it is ℓ -geodesic.*

Proof. (i) Let T be a triangle with all edges tight and let $x \neq y$ be vertices of T . The edge xy is one of the two x - y arcs of T , and every x - y path R of H satisfies $\ell(R) \geq d_\ell(x, y) = \ell(xy)$, so no x - y path is strictly shorter than both arcs. As this holds for all pairs, T is ℓ -geodesic.

(ii) Let $e = b_j b_k$ be non-tight, so $d := d_\ell(b_j, b_k) < \ell(e)$, and let R be a shortest b_j - b_k path, $\ell(R) = d$. Suppose $P_{p;jk}$ is ℓ -geodesic. Its two arcs between b_j and b_k are the edge e and the path $b_j s_p b_k$. Geodesicity forbids $\ell(R) < \ell(e)$ and $\ell(R) < \ell(b_j s_p b_k)$ simultaneously; the first inequality holds, so $\ell(b_j s_p b_k) \leq \ell(R) = d$. Since $b_j s_p b_k$ is itself a b_j - b_k path, also $\ell(b_j s_p b_k) \geq d$; hence $b_j s_p b_k$ is a shortest b_j - b_k path. If $P_{q;jk}$ were also ℓ -geodesic, then likewise $b_j s_q b_k$ would be a shortest b_j - b_k path, and $b_j s_p b_k \neq b_j s_q b_k$ would contradict the uniqueness of shortest paths. $\square$

Theorem 2.2 (Main theorem). *Let G be the graph of Definition 2.1. Then G is a planar 3-connected graph on 8 vertices with 18 edges, and for every assignment $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ some ℓ -geodesic cycle of G is not peripheral.*

Proof. The structural assertions are Lemma 2.4(a) and (c). Suppose, for a contradiction, that some $\ell_0: E(G) \rightarrow \mathbb{R}_{>0}$ makes every ℓ_0 -geodesic cycle of G peripheral. By Lemma 3.2 there is $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ such that every ℓ -geodesic cycle is peripheral and any two shortest paths with the same ends coincide. Fix this ℓ .

Step 1: at most one spoke triangle is not ℓ -geodesic. By Lemma 2.4(c) every ℓ -geodesic cycle is one of the twelve spoke triangles $P_{i;jk}$. By Theorem 3.1 the ℓ -geodesic cycles generate $\mathcal{C}(G)$, which has dimension 11 by Lemma 2.4(a). A set of at most ten vectors spans a space of dimension at most ten, so at least eleven of the twelve spoke triangles are ℓ -geodesic.

Step 2: at most one core edge is not tight. Let $e = b_j b_k$ be a non-tight core edge. By Lemma 3.3(ii) at least one of the two spoke triangles containing e is not ℓ -geodesic; call it $F(e)$. By Lemma 2.4(d) each spoke triangle contains exactly one core edge, so $F(e) \neq F(e')$ for distinct non-tight core edges $e \neq e'$. By Step 1 there is at most one non-geodesic spoke triangle, hence at most one non-tight core edge.

Step 3: a geodesic separating triangle. By Lemma 2.4(e), some core triangle T_i avoids the non-tight core edge if there is one (and any T_i will do if there is none). All three edges of T_i are tight, so T_i is ℓ -geodesic by Lemma 3.3(i). But T_i is not peripheral by Lemma 2.4(c). This contradicts the choice of ℓ , under which every ℓ -geodesic cycle is peripheral.

Hence no ℓ_0 exists: for every $\ell: E(G) \rightarrow \mathbb{R}_{>0}$ some ℓ -geodesic cycle of G is not peripheral. (The argument is indirect and does not say *which* non-peripheral cycle is ℓ -geodesic for a given ℓ ; see Remark 4.2.) $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/geodesic_cycles_and_tuttes_theorem/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The writeup uses exactly the definitions of the source. I retrieved the published Georgakopoulos–Sprüssel paper (arXiv:0911.3999, *Geodetic topological cycles in locally finite graphs*, Electron. J. Combin. 16 (2009) #R144) and checked the wording verbatim:

- §3.1: “A cycle C in G is ℓ -geodetic, if for any $x, y \in V(C)$ there is no x - y path in G of length strictly less than that of each of the two x - y paths on C .” This is precisely the writeup’s reading: C is geodesic iff $d_\ell(x, y) = \min(\text{arc}_1, \text{arc}_2)$ for all pairs of **vertices** of C . Strictness of the inequality (ties do not destroy geodesicity) is used by the writeup and is the source’s convention.
- §5: “Call a cycle in a finite graph peripheral, if it is induced and non-separating.” The writeup’s “induced, and $G - V(C)$ connected” is the same, and is also the definition OPG points to (Wikipedia / Tutte).
- §5, Problem 3, verbatim: “If G is a 3-connected finite graph, is there an assignment of lengths ℓ to the edges of G , such that every ℓ -geodetic cycle is peripheral? We were not able to give an answer to this problem.”

So the writeup refutes the problem as its authors posed it, with the intended quantifier order ($\exists \forall$ geodesic cycles), over arbitrary positive real lengths, and with “cycle” meaning a circuit of the finite graph. This is **not** an “as literally stated” strawman: a reasonable author of the original problem would consider Problem 3 answered in the negative. Note the writeup correctly refrains from claiming anything about Tutte’s theorem itself; only the proposed route to a new proof of it is blocked.

One convention worth recording (see Caveats): GS’s §2 *topological* notion of an ℓ -geodetic **circle** quantifies over all *points* of C , not just vertices. For finite graphs the relevant definition is the §3.1 one (vertices), which is also the one quoted on the OPG page (“for every two vertices x, y on C ”), and which the writeup uses.

B.2 Step-by-step findings

$G: B = \{b_1 \dots b_4\}$ induces a K_4 , $S = \{s_1 \dots s_4\}$ independent, $s_i \sim B \setminus \{b_i\}$. This is the Kleetope of the tetrahedron (triakis-tetrahedron graph), 8 vertices, 18 edges.

#	step	label	note
1	Definition of peripheral (induced + non-separating)	VALID	matches GS §5 verbatim
2	Construction of G ; “insert a vertex in every face of a tetrahedron”	VALID	the planar description agrees with the algebraic one: s_i sits in the face opposite b_i ; $8-18+12=2$
3	G is 3-connected (deleting ≤ 2 vertices leaves ≥ 2 adjacent b ’s, and every surviving s_i keeps a neighbour)	VALID	verified: <code>node_connectivity = 3</code>
4	$\dim_{F_2} C(G) = 18 - 8 + 1 = 11$	VALID	arithmetic re-derived; G connected
5	Every induced cycle of G is a triangle (an induced cycle of length ≥ 4 cannot use any s_i , since its two cycle-neighbours lie in the clique B ; and a ≥ 4 -cycle inside B has a chord)	VALID	brute force: exactly 16 induced cycles, all triangles, out of 239 simple cycles
6	The four core triangles $T_i = G[B \setminus \{b_i\}]$ are induced but not peripheral (s_i is isolated in $G - V(T_i)$)	VALID	verified for all four
7	The twelve triangles $P_{\{i; jk\}} = s_i b_j b_k$ are peripheral	VALID	verified; they are exactly the 12 faces of the planar embedding
8	G has exactly 12 peripheral cycles, each containing exactly one core edge; each core edge lies in exactly two of them	VALID	verified by incidence count (core edge $\rightarrow 2$ faces; spoke edge $\rightarrow 2$ faces)
9	Fact 1: for any positive ℓ the ℓ -geodesic cycles generate $C(G)$ (shortcut Q internally disjoint from C , $C = C_1 \triangle C_2$, both shorter, induct)	VALID	this is GS Theorem 3.1, whose published proof is the same argument; the “every excursion of P can be replaced by a no-longer arc of C ” sentence is the source’s own “it is easy to see” step

#	step	label	note
10	Fact 2: an ℓ with the property can be perturbed to one with the property and unique shortest paths	VALID	the finitely many witness inequalities $\ell(P) < \text{arc1}, \ell(P) < \text{arc2}$ are strict, hence hold on an open neighbourhood; equalities $\ell(P) = \ell(Q)$ between distinct simple paths with the same ends are proper hyperplanes (distinct simple paths have distinct edge sets — the writeup does not say this, a one-line omission), whose union has empty interior. Crucially the writeup only needs “every non-peripheral cycle stays non-geodesic”, which is exactly what the open conditions give; it does not need peripheral cycles to stay geodesic, and it says so.
11	(1) At most one peripheral triangle is non-geodesic (12 spanning vectors in dimension 11)	VALID	verified: the 12 peripheral cycles have GF(2)-rank 11, and every choice of 10 of them has rank exactly 10, so two failures already break generation
12	If a core edge $e = b_j b_k$ is non-tight then $P_{\{p;jk\}}$ and $P_{\{q;jk\}}$ cannot both be geodesic (else both 2-paths would be shortest b_j – b_k paths, contradicting uniqueness)	VALID	re-derived: geodesicity at the pair (b_j, b_k) forces $d = \min(\ell(e), 2\text{-path})$; $d < \ell(e)$ forces $d =$ the 2-path
13	(2) At most one core edge is non-tight	VALID	each non-tight core edge kills a <i>different</i> peripheral triangle by step 8, and (1) allows at most one
14	Some core triangle T_i has all three edges tight	VALID	verified: the minimum number of edges of K_4 meeting all four of its triangles is 2, so one exceptional edge cannot hit every core triangle
15	An all-tight triangle is geodesic	VALID	for each pair, $d(x, y) = \ell(xy) \geq \min(\text{arcs})$, so nothing is strictly shorter than both arcs
16	Contradiction: T_i is geodesic and not peripheral	VALID	closes the proof
17	Closing remarks (planar, simple, 3-connected; self-contained; novelty unchecked)	VALID	all three adjectives verified; the novelty caveat is honest

No step is labelled GAP or ERROR. The only blemishes are stylistic: the internally-disjoint shortcut in Fact 1 and the properness of the hyperplanes in Fact 2 are asserted rather than proved, both being one-liners and the former being the source paper’s own phrasing.

B.3 Reference check

- [GS] arXiv:0911.3999 / *Electron. J. Combin.* **16** (2009) #R144 — fetched and converted to text. Confirmed: the definition of ℓ -geodetic cycle (§3.1) as used; **Theorem 3.1** (“For every finite graph G and every metric representation $(|G|, \ell)$, every cycle C of G can be written as a sum of ℓ -geodetic cycles of length at most $\ell(C)$ ”), which is the writeup’s Fact 1 in a slightly stronger form; the definition of peripheral (§5); and **Problem 3** verbatim, together with “We were not able to give an answer to this problem” and the remark that a positive answer would reprove Tutte’s theorem. Also confirmed: “any assignment of edge lengths yields a metric representation” for finite graphs, so no hidden restriction on ℓ .
- [T] Tutte, “How to draw a graph”, *Proc. LMS* **13** (1963) 743–768 — invoked only as background (peripheral cycles generate the cycle space of a 3-connected graph); used nowhere in the proof. Consistent with the computation (the 12 peripheral cycles of G do have rank 11).
- **OPG entry** — fetched; statement identical, no comment or update announcing a solution or counterexample.
- **Novelty** — web searches (OPG, arXiv, Georgakopoulos’s page, follow-ups on peripheral cycles) surfaced no published counterexample to Problem 3, in agreement with the catalog page’s 2026 literature review. The model itself flags that it did not check novelty. I found no reason to call this ALREADY_KNOWN, but “not found” is weaker than “not there”.

B.4 Computational check

Scripts in verification_astra/scripts/geodesic_cycles_and_tuttes_theorem/ (run with system python3, networkx 3.6.1, z3 4.16.0).

check_graph.py — structure of G : $n = 8, m = 18$, degrees $b_i = 6, s_i = 3$; node_connectivity = 3; planar = True; $\dim C(G) = 11$; 239 simple cycles (16 triangles, 33 C_4 , 60 C_5 , 76 C_6 , 48 C_7 , 6 C_8); **16 induced cycles, all triangles; 12 peripheral cycles**, exactly the $\{b_j, b_k, s_i\}$ triangles; the four $\{b_i, b_j, b_k\}$ triangles are induced but not peripheral. GF(2)-rank of the 12 peripheral cycles = 11; the maximum rank of any 10 of them = 10 (so at most one may fail to be geodesic). All of these match the writeup exactly.

proof_lemmas.py — each peripheral triangle contains exactly one core edge; each core edge and each spoke edge lies in exactly two peripheral triangles; the minimum number of K4-edges meeting all four core triangles is 2.

z3_decide_lean.py — the decisive test. Exact QF_LRA encoding of “ $\exists \ell: E \rightarrow \mathbb{R}_{>0}$ such that every ℓ -geodesic cycle is peripheral”: length variables $1_e > 0$, distance variables with the Bellman characterisation ($d \leq$ every relaxation, and equality attained at some predecessor — for positive lengths this forces the true distance, and true distances always satisfy it, so the encoding cannot spuriously over-constrain), and, for each of the 227 **non-peripheral cycles**, the disjunction over vertex pairs $x, y \in C$ of $d(x, y) < \arccos(1 \wedge d(x, y)) < \arccos(2)$. **Result: unsat**. Since linear real arithmetic is decided exactly (no floating point), this is a machine proof that *no* positive real length assignment on G makes every geodesic cycle peripheral — the writeup’s theorem, obtained independently of its argument.

z3_controls.py — positive controls for that encoding, to rule out an encoding bug producing a vacuous **unsat**. Same code returns **sat** for K4, K5, the cube Q3, the triangular prism, the octahedron and the wheel W4, and in every case the returned rational length vector was re-checked independently with Dijkstra: 0 non-peripheral geodesic cycles. E.g. K4 with $1(01)=1(12)=1(13)=1/8$, $1(02)=1(03)=1(23)=3/8$.

z3_variants.py — robustness of the **unsat** to the definition of “peripheral” on the same graph G : standard (induced + non-separating, 12 cycles) $\rightarrow$ **unsat**; “non-separating only” (120 cycles) $\rightarrow$ **sat**; “induced only” (16 cycles) $\rightarrow$ **sat**. So the counterexample lives exactly on the standard definition, and specifically on the core triangles being induced but separating — which is the mechanism the writeup identifies.

random_search.py — 40 000 random assignments over four distributions (uniform, log-uniform over $e^{\pm 8}$, discrete-ratio, exponential): the number of non-peripheral geodesic cycles was never 0; its distribution was {2: 42, 3: 893, 4: 4784, 5: 9210, 6: 11645, 7: 10075, 8: 3129, 9: 222}, minimum 2. Simulated annealing (20 000 steps on log-lengths) got stuck at 4. In 3 000 samples the number of non-geodesic faces was always $\geq$ the number of non-tight core edges, corroborating step 12–13.

(z3_decide.py, a second encoding that enumerates all simple paths explicitly, is correct but blows up — aborted after ~25 min; it is kept only for reference and is superseded by z3_decide_lean.py.)

B.5 Caveats

1. **Definition sensitivity (not a flaw, but load-bearing)**. The refutation needs “peripheral = induced *and* non-separating”. Under either half alone the same graph admits valid assignments (explicit rational models produced above). The used definition is GS’s own, so this is fine — but any restatement of the result must keep it.
2. **Vertices vs points**. GS’s topological definition for *circles* (§2) quantifies over all points of C ; for finite graphs, Problem 3 lives under the §3.1 vertex definition (also OPG’s phrasing). Step 15 (“an all-tight triangle is geodesic”) is a vertex-level statement and would need re-examination under a point-level reading. This is not the intended reading, but the writeup never remarks on the distinction.
3. **Minimality is neither claimed nor established**. Eight vertices is small but no smaller counterexample is excluded; my controls show K4, K5, Q3, the prism, the octahedron and W4 are *not* counterexamples, which is consistent but not a proof of minimality.
4. **Novelty unverified by the author** (it says so). My searches found nothing, but the topic is niche; a referee should still ask the authors of [GS] whether this example was known to them. If it were known folklore, the verdict would move to ALREADY KNOWN.
5. **Small omissions** in Facts 1–2 (internally-disjoint shortcut; properness of the hyperplanes $\ell(P) = \ell(Q)$), both one-line repairs, neither affecting correctness.
6. **Scope**. The result kills the proposed route to a new proof of Tutte’s theorem via geodesic cycles for this graph; it says nothing against Tutte’s theorem, and the writeup does not pretend otherwise. It also leaves open the natural weakening (which graphs do admit such an ℓ ?) and the infinite version of Problem 3 mentioned in [GS].
7. **Cost of a wrong encoding**. My **unsat** depends on my own model of the problem; I guarded against that with six positive controls and two relaxed variants on the same graph that return **sat**, plus 40 000 randomized trials that agree with **unsat**.

B.6 Referee summary

I set out to break this writeup and could not. The graph is the Kleetope of K4 (tetrahedron with a vertex stacked in each face); every structural claim about it — 3-connectivity, 18 edges, cycle-space dimension 11, “all induced cycles are triangles”, the split into 4 separating core triangles and 12 peripheral face triangles, the one-core-edge-per-face incidence — is verified by brute force and matches the text exactly. The argument is a clean two-branch squeeze: with unique shortest paths (obtained by an open-condition perturbation that only needs the *non-peripheral* cycles to stay non-geodesic, a point the writeup makes explicitly and correctly), each non-tight core edge kills a distinct face, but at most one face may be non-geodesic since 12 face vectors in an 11-dimensional cycle space tolerate only one loss; hence ≥ 5 core edges are tight, hence some core triangle is all-tight, hence geodesic, hence a non-peripheral geodesic cycle. Every cited ingredient checks out against the published source (GS Theorem 3.1 and Problem 3, quoted verbatim), and the interpretation is the intended one rather than a literal-reading strawman. Independently of the argument, an exact z3 decision procedure over linear real arithmetic returns **unsat** for “ $\exists \ell > 0$ with every ℓ -geodesic cycle peripheral” on this graph, while returning verified **sat** models on six control graphs and on the same graph under two weakened notions of peripheral; 40 000 random assignments never got below 2 non-peripheral geodesic cycles. Verdict: CONFIRMED, with the single reservation that novelty against the literature has not been established (nothing found, but the topic is niche), and the two cosmetic gaps noted above.

References

- [1] A. Georgakopoulos and P. Sprüssel, Geodesic topological cycles in locally finite graphs, Electron. J. Combin. 16 (2009), Research Paper R144, doi:10.37236/234; arXiv:0911.3999. Section 3.1 (definition of ℓ -geodesic cycles), Theorem 3.1, Section 5 (definition of peripheral cycles, Problem 3).
- [2] A. Georgakopoulos and P. Sprüssel, Geodesic cycles and Tutte’s Theorem, Open Problem Garden, posted 2007-08-04, http://www.openproblemgarden.org/op/geodesic_cycles_and_tuttes_theorem.
- [3] W. T. Tutte, How to draw a graph, Proc. London Math. Soc. (3) 13 (1963), 743–768.
- [4] R. Diestel, Graph Theory, 5th edition, Graduate Texts in Mathematics 173, Springer, 2017, Section 1.9.

- [5] M. Hamann, Generating the cycle space of planar graphs, [arXiv:1411.6392](#) (2014).